Yash Dave

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Education

Stanford University 2025–2027 (expected)

M.S., Computational and Mathematical Engineering, Data Science Track

GPA: 3.87

Coursework: Theoretical Statistics, AI Measurement Science, AI Reliability, Convex Optimization, Preference Learning, HCI

University of California, Berkeley 2021–2025

B.A., Applied Mathematics (Economics cluster) and Computer Science (Honors); Public Policy Minor

GPA: 3.94

Coursework: ML*, Optimization*, Computer Vision*, Algorithms, Linear Algebra, Probability, Statistical Inference *graduate-level

Publications & Works in Progress

Y. Dave*, S. Truong*, S. Wang†, S. Koyejo†. *The AI Evaluation Ecosystem* . arXiv:2610.09296. AI Measurement Science Workshop, COLM 2026. *equal contribution, †equal advising

M. Hardy, A. Reuel, L. Zhang, J. M. Casabianca, S. Truong, **Y. Dave**, H. Lee, B. W. Domingue, S. Koyejo. *AI Cartography: Mapping the Latent Landscape of AI Benchmark Ecosystems* . ICML 2026.

B. Marks, **Y. Dave**, Z. Wang, H. Chung, R. Patwa, S. Cha, M. Murphy, A. Strang. *Do Generalized-Gamma Scale Mixtures of Normals Fit Large Image Datasets?* . arXiv:2512.17038, under review.

Research Experience

Snowflake **Menlo Park, CA**

Research Intern June–August 2026

- Designed and shipped a user-in-the-loop method that mines sessions to jointly improve memory, skills, hooks, and MCP
- Built multi-turn memory evals and harness-agnostic benchmark infra for Claude Code, Codex, OpenCode, and OpenHands
- Turned real agent sessions into an internal database-engineering benchmark graded by a calibrated LLM judge

Stanford Trustworthy AI Research Lab **Stanford, CA**

Research Assistant (under Prof. Sanmi Koyejo) June 2025–present

- Building an expert-validated generative agent-based model of the AI eval ecosystem to study evaluator and policy design
- Predicting LLM agent success on unseen models and long-horizon coding tasks using explanatory item response theory

Strang Lab **Berkeley, CA**

Research Assistant (under Prof. Alexander Strang) January 2024–September 2025

- Developed a testing pipeline to evaluate realism of a Bayesian mixture model for modeling quasi-sparse image data
- Created a performant library for generating CDFs of prior distributions at scale, finding best-fit parameters, and visualization

Meet Your Future Project (Berkeley Economics and BRAC Uganda) **Berkeley, CA**

Research Apprentice (under PhD Candidate Livia Alfonsi and Prof. Jeremy Magruder) January 2022–October 2022

- Achieved 83% acc. in sentence-level topic classification of a 500+ transcript dataset using NLI-based zero-shot classification
- Performed sentiment analysis and emotion detection using VADER and NRC dictionary, creating visualizations using ggplot2

Talks & Posters

AI Measurement Science Workshop (*Toward Rigorous AI Evaluation*), COLM 2026, San Francisco. *The AI Evaluation Ecosystem* (poster), October 9, 2026. [workshop]

CS Research Lightning Talks, Stanford. *The AI Evaluation Ecosystem*, September 29, 2026.

ICME Research Symposium, Stanford. *The AI Evaluation Ecosystem*, May 19, 2026. [poster]

AI+Science: Accelerating Discovery Conference, Stanford HAI. *The AI Evaluation Ecosystem*, May 5, 2026. [poster] [event]

Stanford Trustworthy AI Research (STAIR) Lab, AI Measurement Science Reading Group. *The AI Evaluation Ecosystem*, April 3, 2026.

Joint Statistical Meeting, Section on Bayesian Statistical Science. *Do Sparsity-Promoting Hierarchical Prior Models Plausibly Model Empirical Image Data*, August 5, 2025. [poster] [event]

UC Berkeley, South Asian Politics (PolSci 145B). *Beliefs about Government Intervention in India*, May 12, 2025. [poster]

MPS Scholars Retreat (3-min SLAM), UC Berkeley. *Do Sparsity-Promoting Hierarchical Prior Models Plausibly Model Empirical Image Data*, April 4, 2025.

Teaching

Course Assistant, CS 229 Machine Learning , Stanford Winter 2026

- Mentored student projects on ML modeling and evaluation; graded assignments and exams; held office hours

Course Assistant, CS 283 Governing Artificial Intelligence , Stanford Autumn 2025

- Developed rubrics and graded technical AI evaluation assignments, policy memos, and philosophy papers

UC Berkeley College of Computing, Data Science and Society **Berkeley, CA**

Teaching Assistant, Data 140 Probability and Data 375 Pedagogy Fall 2024–Spring 2025

- Held two weekly sections; handled Data 375 logistics, worked with professors to design session outlines, and facilitated discussion

Lead Content TA, Data 100 Principles of Data Science Spring 2023–Spring 2024

- Developed and improved content, oversaw beta-testing and directed release of assignments for 1200+ students on JupyterHub
- Incorporated human contexts and ethics (HCE) material with regression/classification projects and discussion worksheets
- Taught 1-2 discussion sections to 20-45 students weekly for 4 semesters; rated 6.55/7 overall effectiveness (n=81)
- Co-authored a public debugging guide and contributed to a data science reference text , reducing office hour tickets by 40%

Honors & Awards

Phi Beta Kappa, UC Berkeley 2025
High Honors in General Scholarship, UC Berkeley 2025
Outstanding GSI Award, UC Berkeley Computing, Data Science and Society 2025
Upsilon Pi Epsilon, Computer Science Honor Society 2025
Dean's List, UC Berkeley 2022–2025
Kevin K. Gong Memorial Scholarship, UC Berkeley EECS 2024
EECS Honors Program, UC Berkeley 2025
World Bank Group Youth Summit Case Challenge, Honorable Mention 2023
Citadel West Coast Regional Datathon, Honorable Mention 2023
EA Berkeley AI Safety Distillation Contest, 3rd Place 2022

Selected Projects

Reliability and Validity of Personalized AI Benchmarks June 2026

- Built a 110-item personalized coding-agent benchmark rated by the user and a cross-lineage LLM judge panel; fit a hierarchical Bayesian Bradley-Terry-Davidson model with generalizability and decision studies to find the cheapest reliable design
- Showed LLM judges are reliable but not valid: they overstate win rates (0.80 vs. the user's 0.57), share correlated errors that do not average out, and give prediction-powered inference no label savings

Direct Preference Optimization applied to Financial Advising December 2025

- Fine-tuned a 4B-parameter model using DPO to automate SEC Marketing Rule compliance review, reducing reasoning tokens by 50% while maintaining expert-level rewrite quality
- Built a privacy-preserving synthetic data pipeline generating 943 preference pairs from 40 expert annotations using a teacher-fixer-breaker framework to train compliant ad reviewers

Beliefs about Govt. Intervention in India May 2025

- Developed a generalized linear regression model to study the relationship between local development and pro-intervention beliefs, controlling for factors like demographics, credit attribution, and government satisfaction

Service, Leadership & Outreach

President, Tech+Social Impact Club May 2023–May 2024

- Co-facilitated student-run course about data ethics, technology and social impact to a total of 100 students over 3 semesters
- Organized a tech help session at Next Village SF, helping 20-30 elders and saving the nonprofit an estimated $1200

Advising Fellow, Matriculate Jan 2022–Aug 2023

- Mentored three first-gen high school juniors throughout the college application process: reviewing essays, scholarships etc.

Data Science and Philosophy through the Spider-verse — Splash @ Stanford (Nov 16, 2025; Apr 11, 2026); Splash @ Berkeley (Apr 13, 2025).

Technical Skills

Python (NumPy, pandas, PyTorch, JAX), R, SQL, C/C++, Go, Java; Git, LaTeX, AWS, CUDA

Languages

English (professional working), Hindi (limited working), Spanish (limited working), Arabic (elementary)